Rates and percents

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

6. A box is 4 cm by 3 cm by 3 cm. Find its volume.

7. 84 ÷ 7

8. Find 10%, 50%, and 75% of 40 counters.

9. Find 140% of 30 points.

10. Find 35% of 80 tiles.

11. 12 shells are 30% of a collection. Find the whole collection.

12. 9 of 12 tiles are green. What percent are green?

13. Omar says 24 is 40% of 9.6 because 0.40 × 24 = 9.6. Explain the error.

14. Find 30% of 100. Name the whole.

15. 120 stickers are 75% of a collection. How many stickers are in the whole collection?

16. 9 of 60 books are mysteries. What percent are mysteries?

17. Find 5% of $27. Give a money answer.

18. A cyclist travels 9 kilometers in 3 hours at steady speed. Find kilometers per hour.

CHALLENGE / Explain your plan and check the result

C1. A $45 bag is reduced by 20% of its price. A $48 bag is reduced by 25% of its price. Find both reductions and final prices. Which final price is lower?

C2. A class has 32 students. Twelve ride bicycles and 8 walk. Find the percent in each group and the percent who do neither. Assume the groups do not overlap.